



ASPIRATING PIPE SYSTEMS



Siemens Product Guide **LPCB Tested**

August 2015

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Material specification and general information

ABS Pressure Pipe Systems

General Information

ABS (Acrylonitrile Butadiene Styrene) is a homogenous material with good chemical resistance and high impact strength.

Other advantageous features are its suitability for use at low temperatures (-40°C) and its ease of jointing.

Standards

Plimat Aspirating Pipe and Fittings are produced under a strict quality control system approved to EN ISO 9001.

Plimat Aspirating Products have been Tested by LPCB to EN 54-20 Clause 5.7, EN 61386-1 Class 1131.

LPCB Test Report No- TE250773

Individual products are in accordance with the appropriate British Standards.

Fittings (inch) BS 5391 Part 1, EN 1452 Part 3, EN 54-20 Clause 5.7, EN 61386-1 Class 1131

Fittings (Metric) Din 8063, Kiwa 549, ISO 727

Pipe BS 5391 EN 61386-1 Class 1131

Adhesive BS 4346 Part 3

Colour

Plimat ABS products are manufactured in Red, White and a Mid Grey Colour.

Temperature

ABS is suitable for use over a wide temperature range from -40°C to 70°C

	Code	Description	Bag Qty
	REDMAI0250	Plain Red ABS 25mm Socket	10
	REDTII0250	Plain Red ABS 25mm 90° Tee	10
	REDBOI0250	Plain Red ABS 25mm Union	10
	REDGYI0250	Plain Red ABS 25mm 45° Elbow	10
	REDGOI0250	Plain Red ABS 25mm 90° Elbow	10
	REDCUI0250	Plain Red ABS 25mm 90° Bend	10
	REDCAI0250	Plain Red ABS 25mm Cap	10
	REDMAB0250	Plain Red ABS 3/4" x 25mm Adaptor	10
	REDADI025A	25mm/10mm Compression Adaptor	10
	REDCTII0250	25mm/10mm Red Capillary Tee	10
	REDADI025T	25mm/10mm Compression Adapter Tee	10
	REDCAP010N	Discreet 10mm Capillary Sampling Point (Natural)	10
	REDAST025I	25mm Red Air Sampling Test Point	1

	Code	Description	Bag Qty
	REDMAP0250	Plain Red ABS 3/4" (27mm)Socket	10
	REDTIP0250	Plain Red ABS 3/4" (27mm)90° Tee	10
	REDBOP0250	Plain Red ABS 3/4"(27mm) Union	10
	REDGY0250	Plain Red ABS 3/4" (27mm)45° Elbow	10
	REDCUP0250	Plain Red ABS 3/4" (27mm)90° Bend	10
	REDCAP0250	Plain Red ABS 3/4" (27mm)Cap	10
	REDMAB0250	Plain Red ABS 3/4"x 25mm Adaptor	10
	REDAMP0250	Plain Red ABS Male 25mm x 3/4" Female Adaptor	10
	REDADP025A	3/4" (27mm)/10mm Compression Adaptor	10
	REDCTIP0250	3/4"/10mm Red Capillary Tee	10
	REDADP025T	3/4"/10mm Compression Adapter Tee	10
	REDCAP010N	Discrete 10mm Capillary Sampling Point (Natural)	10
	REDAST025P	3/4" Red ABS Test Point	1

Code	Description	Bag Qty
 WHTMAI0250	Plain White ABS 25mm Socket	10
 WHTTII0250	Plain White ABS 25mm 90° Tee	10
 WHTBOI0250	Plain White ABS 25mm Union	10
 WHTGYI0250	Plain White ABS 25mm 45° Elbow	10
 WHTGOI0250	Plain White ABS 25mm 90° Elbow	10
 WHTCUI0250	Plain White ABS 25mm 90° Bend	10
 WHTCAI0250	Plain White ABS 25mm Cap	10
 WHTMAB0250	Plain White ABS 3/4"x 25mm Adaptor	10
 WHTCTII0250	25mm x 10mm White Capillary Tee	1
 WHTAST025I	25mm White ABS Air Sampling Test Point	1
 WHTFLXI25A	25mm White Flexible Connector 30cm	1
 WHTFLXI25B	25mm White Flexible Connector 100cm	1
 WHTSGD025I	25mm White Site Glass Condensation Drain	1

ASPIRATING PIPE SYSTEMS

Grey 25mm ABS Fittings

LPCB Tested

Code	Description	Bag Qty
 ABSMAI0250	Plain Grey ABS 25mm Socket	10
 ABSTII0250	Plain Grey ABS 25mm Tee	10
 ABSBOI0250	Plain Grey ABS 25mm Union	10
 ABSGYI0250	Plain Grey ABS 25mm 45° Elbow	10
 ABSGOI0250	Plain Grey ABS 25mm 90° Elbow	10
 ABSCUI0250	Plain Grey ABS 25mm 90° Bend	10
 ABSCAI0250	Plain Grey ABS 25mm Cap	10
 REDTII010	10mm 3 Way Compression Tee	1
 ABSMAB0250	Plain Grey ABS 3/4" x 25mm Adaptor	1

Code	Description	Bag Qty
 ABSMAP0250	Plain Grey ABS 3/4" (27mm)Socket	10
 ABSTIP0250	Plain Grey ABS 3/4" (27mm)90° Tee	10
 ABSBOP0250	Plain Grey ABS 3/4"(27mm) Union	10
 ABSGYP0250	Plain Grey ABS 3/4" (27mm)45° Elbow	10
 ABSGOP0250	Plain Grey ABS 3/4" (27mm)90° Elbow	10
 ABSCUP0250	Plain Grey ABS 3/4" (27mm)90° Bend	10
 ABSCAP0250	Plain Grey ABS 3/4" (27mm)Cap	10
 REDTII010	10mm 3 Way Compression Tee	1
 ABSMAB0250	Plain Grey ABS 3/4" x 25mm Adaptor	1
 GRYFLXI025A	Grey Flexible Connector 30cm New	1
 GRYFLXI025B	Grey Flexible Connector 100cm New	1

	Code	Description	Bag Qty
	REDCLI0250	Red 25mm ABS Pipe 3 Metre Lengths	20
	WHTCLI0250	White 25mm ABS Pipe 3 Metre Lengths	20
	ABSCLI0250	Grey 25mm ABS Pipe 3 Metre Lengths	20
	REDCLE0250	Red 3/4" ABS Pipe 3 Metre Lengths	20
	ABSCLE0250	Grey 3/4" ABS Pipe 6 Metre Lengths	20
	REDLDP010R	Red 10mm Capillary Pipe 100 metre	1
	REDLDP010N	Natural 10mm Capillary Pipe 100 metre	1
	REDLDP006N	Natural 6mm Capillary Pipe 100 metre	1



ASPIRATING PIPE SYSTEMS

Capillary Kits 25mm - 10mm

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Code	Description	Bag Qty
 REDCKI025F	25mm Red Capillary Kit with Flush Air Sample Point	1
 REDCKI025C	25mm Red Capillary Kit with Conical Air Sampling Point	1
 REDCKT025C	25mm Red Capillary Kit Test Point with Cap	1
 WHTCKI025F	25mm Natural Capillary Kit with Flush Air Sample Point	1
 WHTCKI025C	25mm Natural Capillary Kit with Conical Air Sampling Point	1
 WHTCKT025C	25mm Natural Capillary Kit Test Point with Cap	1
 REDCKIT25F	25mm Red Capillary Kit with Flush Air Sample Point / Tee	1
 REDCKIT25C	25mm Red Capillary Kit with Conical Air Sampling Point / Tee	1
 REDCKT025T	25mm Red Capillary Kit Test Point with Cap / Tee	1
 WHTCKI025FT	25mm Natural Capillary Kit with Flush Air Sample Point / Tee	1
 WHTCKI025CT	25mm Natural Capillary Kit with Conical Air Sampling Point / Tee	1
 WHTCKT025T	25mm Natural Capillary Kit Test Point with Cap / Tee	1
 REDCKI025D	25mm Red Capillary Kit with Discreet End Cap	1
 WHTCKI025D	25mm Natural Capillary Kit with Discreet End Cap	1



ASPIRATING PIPE SYSTEMS

Capillary Kits 3 / 4" (27mm) - 10mm

LPCB Tested

Code	Description	Bag Qty
 REDCKP025F	3/4" (27mm) Red Capillary Kit with Flush Air Sample Point	1
 REDCKP025C	3/4" (27mm) Red Capillary Kit with Conical Air Sampling Point	1
 REDCKTP025C	3/4" (27mm) Red Capillary Kit Test Point with Cap	1
 REDCKPT025F	3/4" (27mm) Red Capillary Kit with Flush Air Sample Point/Tee	1
 REDCKPT025C	3/4" (27mm) Red Capillary Kit with Conical Air Sample Point/Tee	1
 REDCKTP025T	3/4" (27mm) Red Capillary Kit Test Point with Cap/Tee	1
 REDCKPT025D	3/4" (27mm) Red Capillary Kit Discreet / with Tee	1

ASPIRATING PIPE SYSTEMS

Clips and Bracketry 25mm - 3/4"

LPCB Tested

	Code	Description	Bag Qty
	REDPCR0250	Red 25mm x 3/4" Pipe Clip Quantity	Box 500
	REDPCR025A	Red 25mm x 3/4" Pipe Clip Bagged	100
	WHTPCR0250	White 25mm x 3/4" Pipe Clip Quantity	Box 500
	WHTPCR025A	White 25mm x 3/4" Pipe Clip Bagged	100
	ACCPCP0250	Grey Pipe Clips with safety strap	1
	REDSTD025A	M6 Stand off with clip and nut (60mm)	1
	CR225	25 - 30mm Pipe Clips 2-4mm Flange	100
	CR425	25 - 30mm Pipe Clips 4 - 7mm Flange	100
	CR825	25 - 30mm Pipe Clips 8 - 12mm Flange	100
	VFCL1100	25 - 32mm Pipe Clips 1 - 5mm Flange	100
	VFCL2-100	25 - 32mm Pipe Clips 5 - 7mm Flange	100
	ACCPCN025A	M6 Nut (for fitting clip to the end of threaded rod)	10
	ACCPCN025B	M8 Nut (for fitting clip to the end of threaded rod)	10
	ACCPCA025A	M6 Rod adaptor (for fitting clip to side of rod)	10
	ACCPCA025B	M8 Rod adaptor (for fitting clip to side of rod)	10
	ACCRPA025A	M6 Raw Plug Adaptor (for fixing clip to wall)	100

Code	Description	Bag Qty
 ACCREM0250	ABS Cement 250ml Tin	1
 REDASP025F	Flush Air Sampling Point (White with Label)	1
 REDASP025C	Conical Air Sampling Point (White with Label)	1
 REDAST025C	Air Sampling Test Point with Cap	1
 REDLAB001	Air sampling labels 23mm /4mm hole	100
 REDLABSIEMENS 2	Siemens Wrap Around Label	100
 REDFLXI25A	25mm Flexible Connector 30cm	1
 REDFLXI25B	25mm Flexible Connector 100cm	1
 REDSGD025I R	25mm Sight Glass Condensation Drain	1
 REDSGD025P	3/4" Site Glass Condensation Drain	1
 REDFLXP25A	3/4" (27mm) Flexible Connector 30cm	1
 REDFLXP25B	3/4" (27mm) Flexible Connector 100cm	1
 REDCUT0250	Pipe Cutter 0 – 42mm	1
 REDCUT025E	Economy Pipe Cutter 0 – 42mm	1
 REDALA025I	25mm x 3/8" BSP Female Air- Line Adaptor	1



ASPIRATING PIPE SYSTEMS

Accessories

LPCB Tested

Code	Description	Bag Qty
 RED22P020I	25mm Compact 2 Way Ball Valve	1
 RED22P0200	3/4" Compact 2 Way Ball Valve	1
 RED30P020I	25mm Compact Check Valve (Air Release)	1
 RED30P0200	3/4" Compact Check Valve (Air Release)	1
 RED22P025I	25mm Large 2 Way Ball Valve	1
 RED22P0250	3/4" Large 2 Way Ball Valve	1
 RED30P025I	25mm Large Check Valve (Air Release)	1
 RED30P0250	3/4" Large Check Valve (Air Release)	1
 RED30AF025I	25mm Large Air Filter	1
 RED30AF0250	3/4" Large Air Filter	1
 PVC30P020I	25mm PVC Compact Check Valve (Air Release)	1
 ABSPQRAV0250	ABS Quick Release with valve 25mm	1



ASPIRATING PIPE SYSTEMS

Accessories

LPCB Tested

Code	Description	Bag Qty
 REDIEJ025I	25mm Inline Expansion Socket 100mm	1
 REDFCR0320	25mm - 32mm Intumescent Fire Collar	1
 REDSGVD025I	25mm x 12" Condensation Drain with D/Union Ball Valve	1
 REDSGVD0250	3/4" x 12" Condensation Drain with D/Union Ball Valve	1
 REDSDCT025I	25mm Self Draining Condensation Trap	1
 REDSDCT0250	3/4" Self Draining Condensation Trap	1
 REDSGVD040I	25mm Large Condensation Drain with Pipe Clips	1
 REDSM12	Smoke Matches	1
 REDSP6	Smoke Pellets	1
 REDSP1	Smoke Pen	1
 REDCT100	Cable Ties	100
 REDSTW100	Smoke Test Wire	1
 EM50520604	Beam Clip with Attachment 2 -4mm	1
 EM50520609	Beam Clip with Attachment 5 - 9mm	1
EM50520616	Beam Clip with Attachment 10 -16mm	1
EM50520620	Beam Clip with Attachment 17 - 20mm	1
EM50520630	Beam Clip with Attachment 23 - 30mm	1

Technical

Jointing Guidelines

Plimat Solvent cement is specially formulated to chemically weld pipes and fittings together. The solvent cement chemically melts the two surfaces to be joined, so that when they are fitted together they form a homogenous mass, which then cures to form a weld. Note that this is not a glued joint. It is therefore important to choose the correct type of adhesive as another type may be detrimental to the integrity of the system.

- 1) Cut the pipe at right angles to its axis, and to the required length using the correct cutting shears.
- 2) Dry fit the pipe to the socket of the fittings. When the pipe is fully home in the socket, draw a line around the pipe at the edge of the socket. Where this is not possible (perhaps on larger fittings) measure the socket depth and draw a line at the corresponding point along the pipe. This will give a visual indication, to ensure that the pipe is fully pushed home in the socket.
- 3) Apply the solvent cement with a suitably sized brush or the brush provided in the Plimat adhesive lid. Ensure that the area of the pipe up to the visual indicator is completely covered with an even layer of cement. This part of the operation must be done quickly and neatly, as the solvent must still be wet when the pipe and fitting is pushed together.
- 4) Push the pipe and fittings together and hold in place for up to 30 seconds. When the joint is made, a bead of solvent cement will form around the outer joint of the pipe and socket. This excess cement should be wiped away leaving the outer part of the joint clean.

Jointing, "Do Not"s

- a) Make joints in rain or wet conditions
- b) Use dirty brushes or cleaning rags, which are dirty or oily.
- c) Use the same brushes with different solvent cements.
- d) Dilute or thin solvent cements with cleaner.
- e) Leave solvent cement tins open. The contents will evaporate and the cement performance will be weakened.
- f) Use near naked lights, or smoke whilst jointing. Solvents are highly inflammable.
- g) Make joints in a confined space. Solvents emit hazardous vapours, which are dangerous.

Joints per Litre of Cement

As a rule of thumb we estimate that 170 joints can be made per 250ml of Plimat solvent cement when jointing either 25mm or ¾" ABS fittings and pipe.

Expansion and Contraction

Expansion or contraction of plastic pipe is caused by temperature change occurring within the pipe wall. When the operating temperature of a pipe is greater than when it was installed, then the pipe will expand. If the operating temperature is lower, then it will contract.

There are two factors to consider when calculating expansions or contractions in pipes.

- 1) ambient temperature of the (air temp) environment when installing the pipe
- 2) change of temperature of pipe contents or environment

Any change of the above factors will affect the mid-wall temperature of the pipe thus causing either expansion or contraction.

Please note that most pipework systems are installed between 5°C and 25 °C.

Calculation Expansion/Contraction

- 1) The change in length due to contraction or expansion in a pipe system is determined by the following formula:

$$\Delta L = \Delta T \times L \times \square$$

Where	ΔL	Expansion (ΔL_e) or contraction (ΔL_c) in mm
	ΔT	Difference in temperature between the installation and the operating temperatures in °C (= $T_{\text{operate}} - T_{\text{install}}$)
	L	Length of pipe when installed
	$\square$	Relevant coefficient of expansion

Example:

Find the expansion and contraction on a 25mm diameter ABS pipe system, installed at 10°C. The maximum and minimum operating temperatures are 30°C and 8 °C respectively. The overall length of the installation is 30m.

Step 1) Calculate temperature change for expansion and contraction:

$$\Delta T = 30 - 10 = +20 \text{ }^{\circ}\text{C}.$$

$$\Delta T = 8 - 10 = -2 \text{ }^{\circ}\text{C}.$$

Step 2) Now calculate expansion and contraction,

$$\text{Expansion } \Delta L_e = \Delta T \times L \times \square\square\square 20 \times 30 \times 0.100 = 60\text{mm}$$

$$\text{Contraction } \Delta L_c = \Delta T \times L \times \square\square\square -2 \times 30 \times 0.100 = -6\text{mm}$$

Step 3) in order to provide for the correct solution, it is necessary to take the greater value, regardless whether it is due to expansion or contraction.
i.e. $\Delta L = 60\text{mm}.$

How to Allow for Expansion or Contraction

The change of length in a pipe system, whether it is expansion or contraction, will require compensation, so that any stresses generated by the change will not cause damage to the system, this can be done in two ways, either by installing

- 1) An Expansion Loop.
- 2) An Expansion Joint.
- 3) Flexible arms.

1) Expansion Loops

These are designed to compensate for linear expansion or contraction within a pipe system. They are an option when facing a large amount of movement due to expansion or contraction. For large amounts of movement we have flexible connectors; solvent weld or mechanical joint connect one end to the pipe where you wish the expansion to be taken up and form a loop before connecting to the other end to the pipe system, the loop must be supported in the middle with a pipe clip, cable clip or other. Make sure the flexible loop will give the pipe enough movement for expansion and contraction and not put the flexible joints under stress. Suitable for use within a temperature range of -20°C to 50°C. See Data Sheet for these products

1) Expansion Joints

In line expansion sockets are a compact solution to allow for 70mm of travel. They have a fixed bracket which is clamped down securely and enables it to be screwed onto M8 or M10 threaded bar. The pipe is pushed into the expansion socket at both ends and must pass the o-ring on each end of the expansion joint. For contraction requirements the pipes can meet in the middle, and for expansion purposes the pipe from each side has to pass the o-ring sufficiently as to not pull out, whilst the space between the pipes inside the expansion socket will allow for the linear expansion of the pipes upto 70mm.

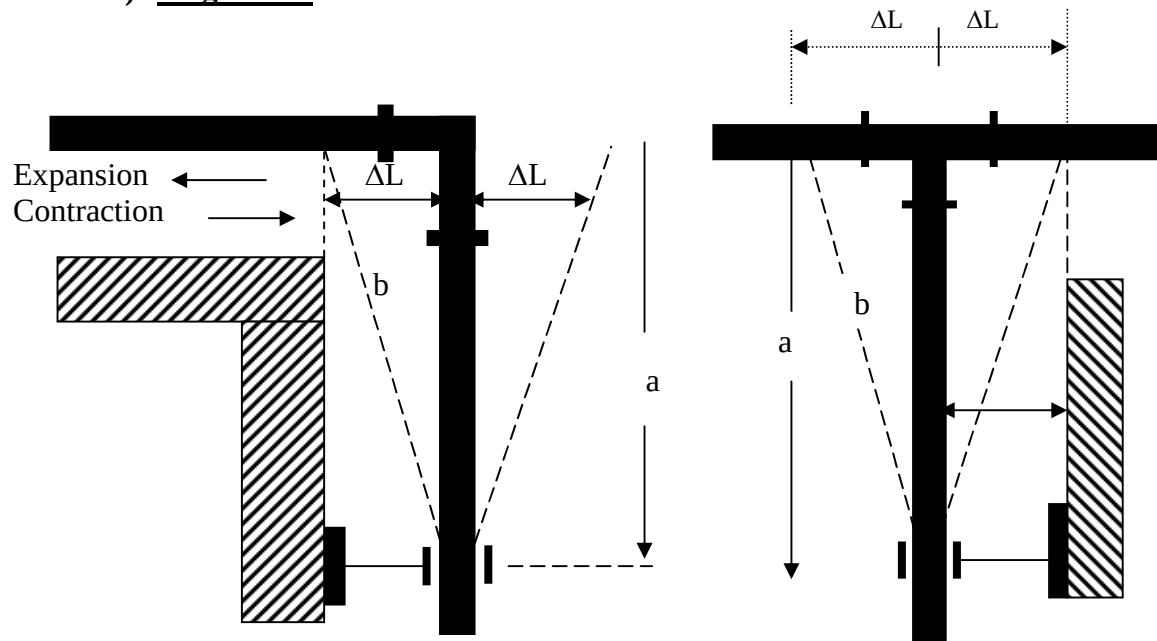
2) Flexible Arms

Flexible arms are simple and relatively inexpensive to install. The flexibility of plastics permits expansion or contraction to be compensated for, by means of, either directional change within a pipe system, or by the installation of expansion loops consisting of two flexible arms.

The length of the flexible arm is governed by the pipe diameter, and the amount of expansion or contraction that requires compensation.

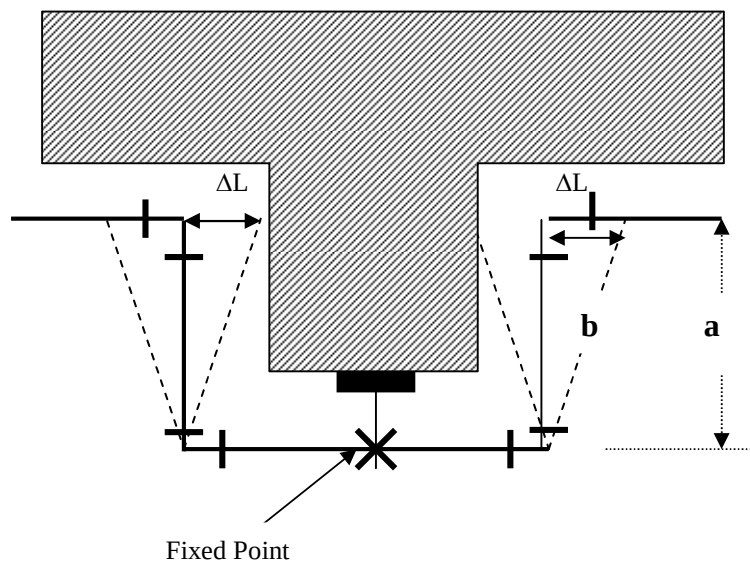
The following sketches show examples of flexible arms that can be utilised within a system.

1) Single Arm



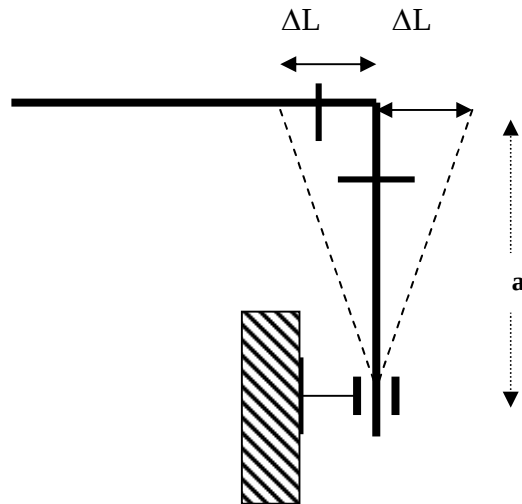
- a flexible arm length
- b distance from pipe OD to wall = $\Delta L + 10\text{mm}$
- ΔL expansion/contraction length.

2) Double Arm (expansion loop)



How to find the flexible arm (a) length.

Single Arm



To calculate the length of a flexible arm (a) the following formula can be used.

$$a = \left(\sqrt{D \times \Delta L} \right) \times C$$

Where a = flexible arm length

d = pipe outside diameter

ΔL = expansion/contraction (mm)

C = constant for material, see Table 7.2

Table 7.2 Thermoplastic material constant C, used in calculating Flexible arm length

Thermoplastic Material	Constant C
PVC-u	33.5
ABS	32.7
PP	30.0
PE	26.0

Example 1

A 40mm ABS pipe has expanded in length 22mm.

What is the length required for the flexible arm.

Solution :

$$a = \left(\sqrt{D \times \Delta L} \right) \times C = \left(\sqrt{40 \times 22} \right) \times 32.7$$

$$a = 970 \text{ mm}$$

Example 2

Use the same parameters as example 1, and apply the double arm equation.

Solution :

The equation for a double arm expansion loop is as follows:

$$a = \left(\sqrt{D \times \frac{\Delta L}{2}} \right) \times C$$

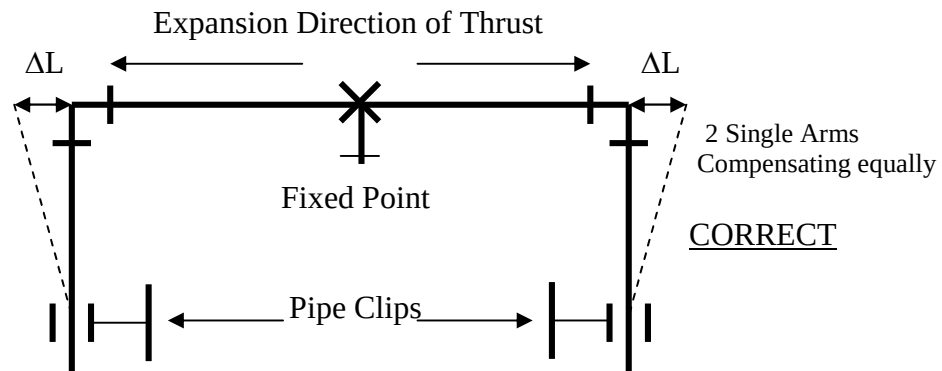
The equation is shown in this form, as the overall expansion (ΔL) is now compensated for over 2 flexible arms.

$$a = \left(\sqrt{40 \times \frac{22}{2}} \right) \times 32.7 = 685.9 \text{ mm}$$

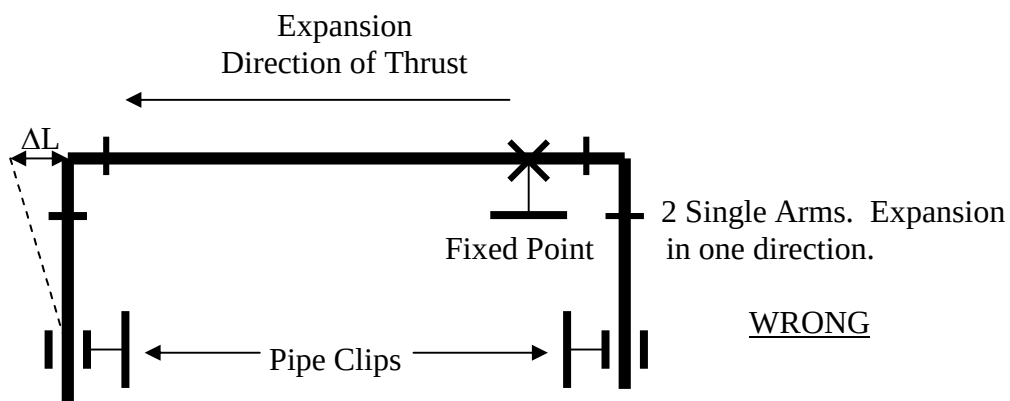
round up to 686 mm

Installation of Expansion Arms Single and Double (Loop)

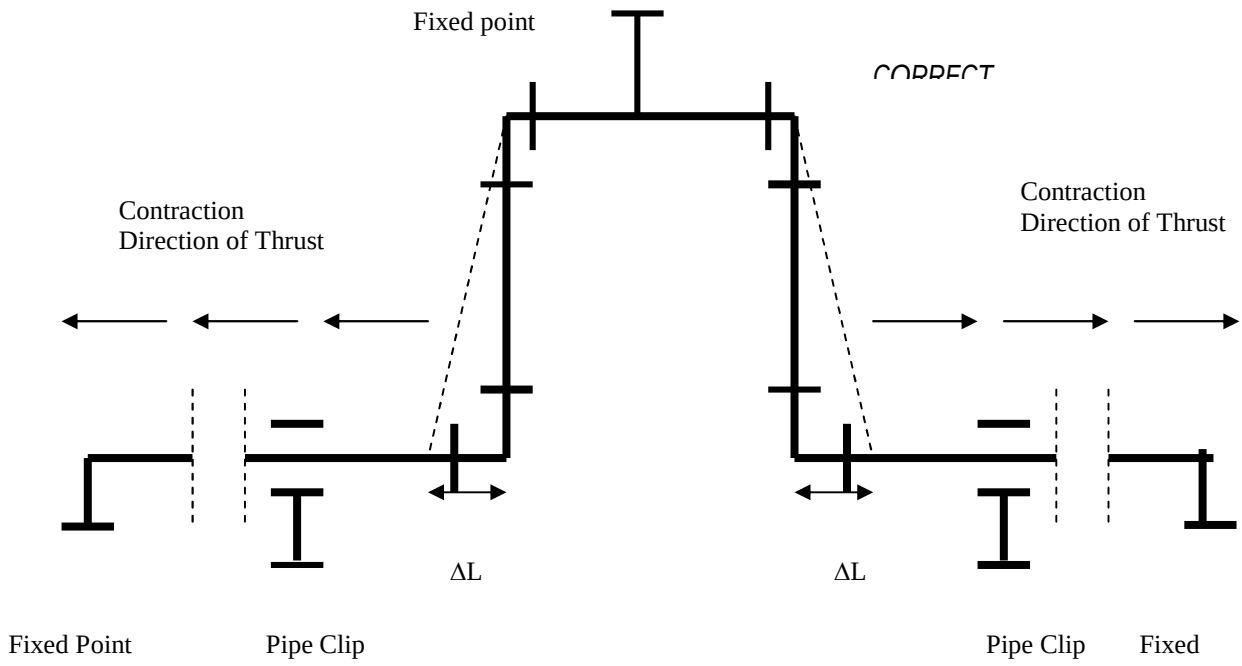
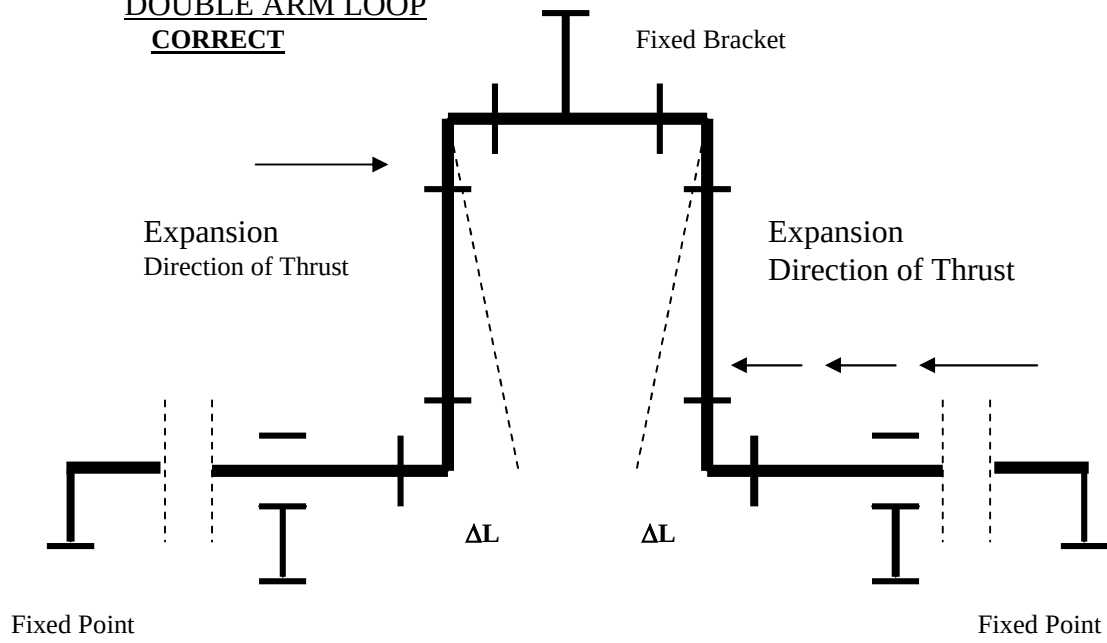
In order to effectively utilise flexible arms, it is necessary to channel any change of length towards that arm, regardless of whether the change is expansion or contraction. This can be achieved by the use of strategically placed fixed brackets, anchoring the pipe firmly.



The bracket has been placed to ensure even distribution of length change, either side of the fixed point. It is not advisable to place the bracket so that the distribution is uneven.



DOUBLE ARM LOOP CORRECT



Pre-Stressing Flexible Arms

Sometimes changes of length (ΔL) can only be channelled in one direction, possibly due to a flexible section having to operate in a confined space. When this occurs, the flexible arm can be pre-stressed. This achieves the following:

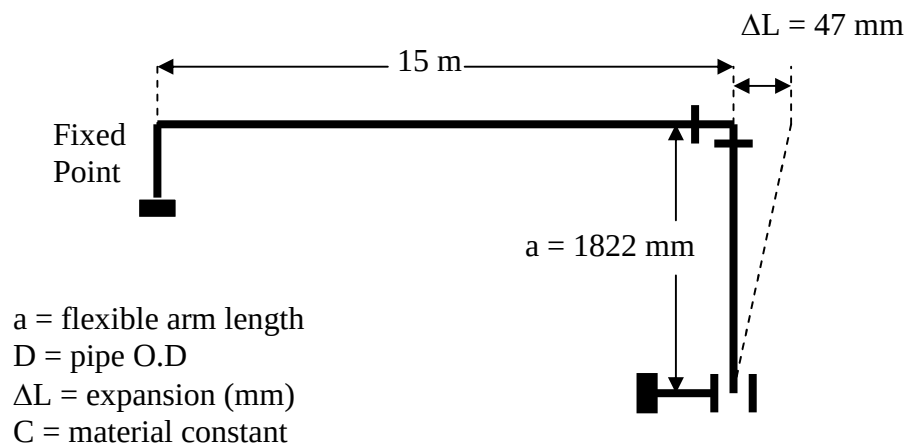
- 1) The flexible arm can be reduced in length
- 2) The flexible arm will straighten under working conditions thus relieving a large amount of stress.
- 3) The installation will look better when in service.

The following examples will demonstrate the above.

Example 3

Non Pre-Stressed Section

A 15 metre length of 63mm PVC-u pipe with a working temperature of 40 °C has expanded (ΔL) 47mm lineally.



$$\therefore a = (\sqrt{D \times \Delta L}) \times C = (\sqrt{63 \times 47}) \times 33.5$$

$$a = 54.4 \times 33.5 = 1822 \text{ mm}$$

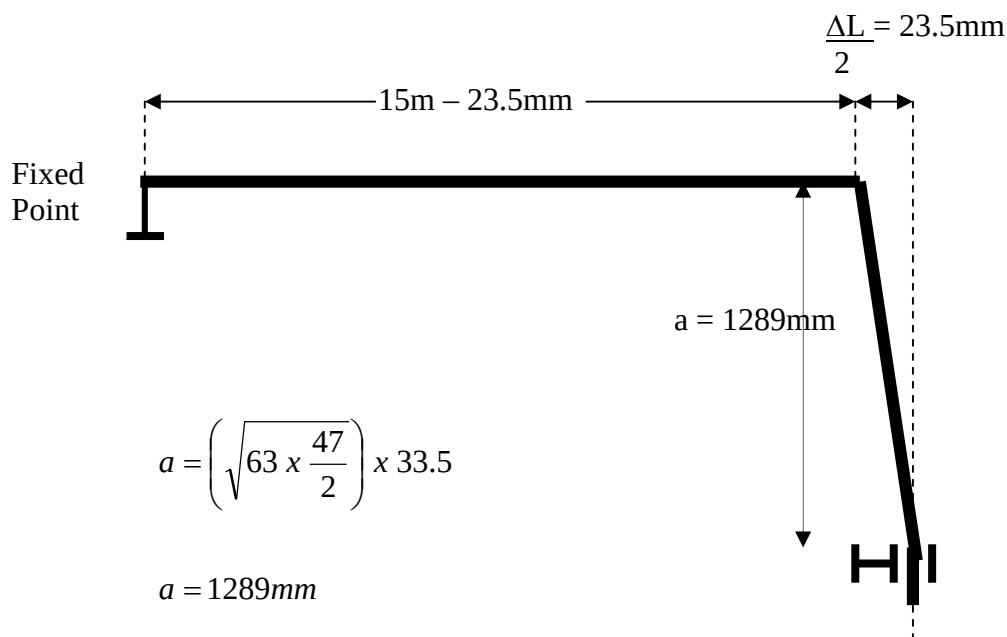
Example 4

Pre-Stressed Section

Using the previous example we shall now pre-stress the same section. To achieve this the overall expansion of 47mm must be taken into account, and reduced by half.

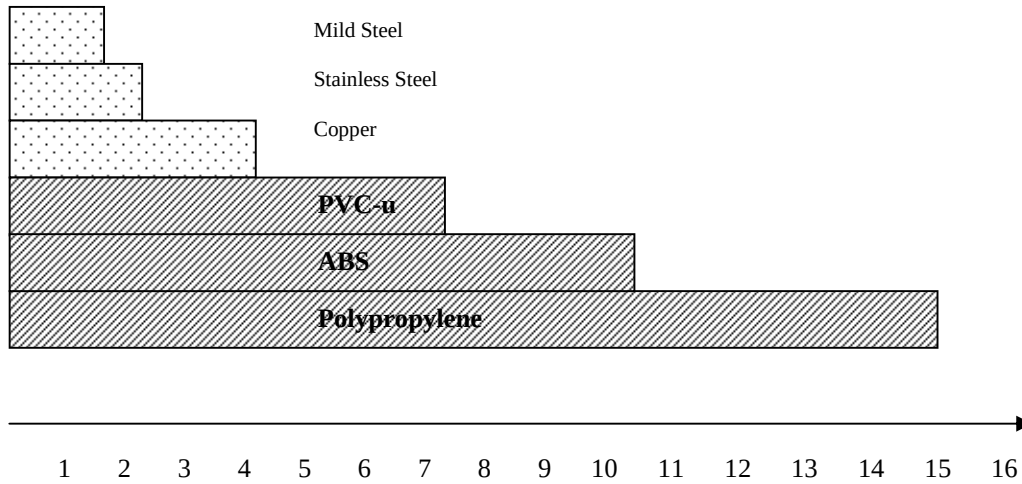
i.e. $\frac{\Delta L}{2} = \frac{47}{2} = 23.5\text{mm}$

What this means is, that if we cut 23.5mm off the 15m length of pipe, we will only need to allow for the remaining 23.5mm of expansion.



By pre-stressing the flexible arm it has been possible to reduce the arm length, and the pipe can now expand and allow (a) the flexible arm to straighten and to relieve the pre-stress.

Coefficient of Linear Expansion for Plastics



Comparison of thermal expansion of plastics and metals

Coefficient of Linear Expansion for Plastics

Material	Coefficient δ $\times (10^{-5} \text{ m/m}^\circ\text{C})$	Length/Temp $^\circ\text{C}$ Equipment mm/ m°C
PVC-u	7.8	0.078
ABS	10.1	0.100
P.P	15.0	0.150
P.E.	20.0	0.200

The meaning of the above coefficients are for example: ABS will expand 0.100mm per metre, for every 10°C raised in mid-wall temp above the installation temperature.

Expansion in mm from 1°C to 20°C for 1 metre Length Pipe

$^\circ\text{C}$	ABS	
1	0.100	For the temperature range not on the chart, add the factors. i.e. for ABS @ 37°C Add $20^\circ\text{C} = 2.000$ <u>$17^\circ\text{C} = 1.700$</u> <u>$37^\circ\text{C} = 3.700$</u>
2	0.200	
3	0.300	
4	0.400	
5	0.500	
6	0.600	
7	0.700	
8	0.800	
9	0.900	
10	1.000	
11	1.100	
12	1.200	
13	1.300	
14	1.400	
15	1.500	
16	1.600	
17	1.700	
18	1.800	
19	1.900	
20	2.000	

Clips and Bracketing

Pipe brackets need to be made with the inside diameter of the bracket marginally larger than that of the pipe outer diameter. This allows for free lineal movement of the pipe, and avoids inhibiting expansion or contraction. They should also be smooth, to avoid damage to the outer surface of the pipe.

Plimat's plastic pipe clips meet all these requirements, and are strong, durable against temperature, Ultra Violet light and can also be used in corrosive or otherwise adverse environmental conditions.

Plimat pipe clips are adaptable and can be fixed in the normal way or be screwed onto M6 or M8 threaded bar by inserting the respective nut into the side of the clip. They can also be fitted to the side of M6 or M8 threaded bar by utilising a rod adaptor. For an example of this please look to the photograph in the bottom left corner of the contents page

Bracket Spacing Intervals

Plastic pipe lines require regular support, and the spacing of clips or brackets depends on the pipe used, temperature, and density of the liquid carried.

The following tables show the centre to centre measurement between brackets.

ABS Pipe 15 Bar (217 Psi) Class E, Inch Sizes

Pipe Size		Bracket Spacing in Metres					
Inch	Metric	20°C	30°C	40°C	50°C	60°C	70°C
$\frac{3}{8}$		0.80	0.75	0.65	0.60	0.50	0.40
$\frac{1}{2}$	20	0.90	0.80	0.75	0.65	0.55	0.50
$\frac{3}{4}$	25	1.00	0.95	0.85	0.75	0.70	0.60
1	32	1.10	1.00	0.95	0.80	0.75	0.65
$1\frac{1}{4}$	40	1.20	1.10	1.00	0.90	0.80	0.70
$1\frac{1}{2}$	50	1.25	1.20	1.10	0.95	0.85	0.75
2	63	1.40	1.30	1.20	1.00	0.90	0.80
$2\frac{1}{2}$	75	1.50	1.35	1.25	1.15	1.00	0.85
3	90	1.60	1.45	1.35	1.20	1.05	0.90
4	110	1.80	1.65	1.55	1.35	1.20	1.00
5	140	2.00	1.80	1.70	1.50	1.30	1.10
6	160	2.10	1.90	1.80	1.60	1.40	1.20

PLIMAT - Trading Terms and Conditions

1 GENERAL

"Seller" shall mean Plimat (UK) Limited, Registered in England under number 03048432. "Buyer" shall mean any company, organisation or individual to whom a quotation is offered, or whose order is accepted by the Seller.

2 CONDITIONS

All offers, quotations, estimates, acceptances and contracts are subject to these Conditions of Business and any terms or conditions which any other person shall seek to impose or make part of any contract shall, so far as is inconsistent with these Conditions of Business, not apply unless expressly agreed by the Seller in writing. The headings in these conditions are for convenience only and shall not affect their interpretation unless expressly agreed by the Seller in writing. The headings in these conditions are for convenience only and shall not affect their interpretation.

3 ELECTRONIC TRADING

If the Seller and Buyer agree that electronic trading between them shall be a basis for ordering processing and invoicing then these terms and conditions shall apply subject to any special terms and conditions terms which are specific to electronic trading and which have been agreed by the parties in writing. Electronic orders shall be valid if all the information agreed between the Buyer and Seller as being required is properly set out in the agreed format and the order is transmitted by the Buyer to the Seller by reference to the correct identification code and is received by the Seller when collecting its electronic mail from the relevant system.

4 QUOTATION

- (a) A quotation by the Seller does not constitute an offer and the Seller reserves the right to withdraw or amend the same at any time prior to the Seller acceptance of the Purchaser's order.
- (b) Unless stated otherwise, all quotations and published price lists are ex works, exclusive of VAT and shall remain valid for 30 days or such a period as may be quoted but nevertheless the Seller may amend or withdraw any quotation by written or oral notice. Quotations may be varied if the Buyer makes variations in his specifications.

5 PRICES

- (a) The Seller will endeavour to ensure that all prices on display/provided to Buyer are correct and up to date. However, should a Buyer place an Order using an incorrect price then the Buyer agrees that the Seller may substitute the incorrect price set out in the Order for the correct price standard price, as appropriate and charge accordingly.
- (b) The Seller shall be entitled at any time by giving notice in Writing, before or after final invoicing to make a reasonable adjustment to the price in the event of any alteration in quantity, design or specification requested by the Buyer.
- (c) The Seller reserves the right at any time prior to delivery by giving notice in Writing to increase the price if there is any increase in cost of materials, labour, transport or utilities or if the costs of the Seller are increased by any other factor beyond the reasonable control of the Seller.

6 DELIVERY

- (a) Except where stated to the contrary in the contract, delivery shall be made as follows:
 - (i) where the Buyer provides the transport, delivery shall be made ex the Seller's works;
 - (ii) where the Seller provides the transport, delivery shall be made to the premises of the Buyer, or the premises of the Buyer's customer or works site if the Buyer has requested delivery to be so made but where the Buyer has made such a request the Seller will make a first delivery to the Buyer's customer or works site as so much of goods as is available for that delivery but subsequent deliveries will be made to the premises of the Buyer.
- (b) Unless otherwise expressly agreed in Writing the Seller may effect delivery in one or more instalments. Where delivery is effected by instalments, the instalment shall be treated as a separate contract governed by these conditions. No delay in the delivery of any instalment of Products or any defect therein shall entitle the Buyer to terminate the remainder of the contract.
- (c) Where goods are sent FOB the Seller's responsibility shall cease when the goods are placed on board ship or aircraft without the need for the Seller to give notice to the Buyer and the provisions of Section 32(3) of the Sale of Goods Act 1979 shall not apply.

7 RISK AND TITLE

- (a) Risk of damage to or loss of the Products shall pass to the Buyer upon delivery and the Buyer is then solely responsible for all loss damage or deterioration to the Products.
- (b) Title to the Products shall not pass to the Buyer until either:
 - (i) the Seller has received in cash or cleared funds all monies payable (whether or not due) to the Seller under this and any other contracts whenever made between the Seller and the Buyer including contracts made after this contract; or
 - (ii) when the Seller serves on the Buyer notice in Writing specifying that title in the Products or any part thereof has passed.
- (c) Until title has passed to the Buyer the Seller may require the Buyer to deliver up to the Seller all Products in respect of which the Seller has title and if the Buyer fails to do so forthwith the Buyers officers, employees, representatives or agents shall be entitled to enter upon any premises where such Products are kept for the purposes of recovering the same.

- (d) Until title the Products has passed to the Buyer pursuant to these conditions it shall possess the Products as fiduciary agent and bailee of the Seller and shall store the Products separately from other goods not owned by the Seller and shall ensure that they are fully insured on all risks basis and clearly identifiable as belonging to the Seller and the Seller shall be entitled to enter upon any premises where such Products are kept for the purpose of satisfying itself that this condition is being complied with by the Buyer.

8 TERMS OF PAYMENT

- (a) Unless otherwise agree by the Seller in Writing, the Buyer shall make payment by the last day of the month following the month of invoice and the Seller shall be entitled to issue invoices in the month in which the Products are delivered or would have been delivered, save for postponement or delay otherwise than due to default on the part of the Seller. Time for payment of the price is of the essence of the contract.
- (b) No disputes arising under this contract shall serve to permit payment by the Buyer of sums due to the Seller to be delayed nor shall disputes interfere with prompt payment in full. The Buyer shall not be entitled to make any deduction from or set off against any sums owing to the Seller by reason of any such dispute or at all.
- (c) In the event of default in payment by the Buyer and Seller shall be entitled, without prejudice to any other right or remedy; to suspend without notice all further deliveries on this or any other contract between the Seller and the Buyer.
- (i) to charge interest on a daily basis (after as well as before judgement) on any amount outstanding at the rate of 4% above the Base Rate of HSBC Bank from time to time; and/or
- (iii) to serve notice on the Buyer requiring immediate payment for all goods supplied by the Seller under this and all other contracts between them whether or not payment is otherwise due or invoiced.

9 SHORTAGES AND DEFECTS APPARENT ON DELIVERY

- (a) It shall be the responsibility of the Buyer to inspect or arrange for an inspection of the goods on delivery whether the goods are delivered to the Buyer's premises or to the premises of the Buyer's customer or to a works site. If no such inspection is made the Buyer shall be deemed to have accepted the goods.
- (b) The Buyer shall have no claim for shortages or defects apparent on inspection unless:
 - (i) a written complaint is made to the Seller within three days of receipt of the goods specifying the shortage or defect; and
 - (ii) the Seller is within seven days of receipt of the complaint given an opportunity to investigate the complaint before any use is made of the goods.
- (c) If a complaint is not made to the Seller as herein provided then in respect of such shortages or defects the goods shall be deemed to be in all respects in accordance with the contract and the Buyer shall be bound to pay for the same accordingly.

10 CLAIMS FOR DEFECTS NOT APPARENT ON INSPECTION

- (a) The Buyer shall have no claim for defects not apparent on inspection unless the Seller is notified in Writing of defective workmanship or materials within twelve months from delivery of the goods. Provided that the goods have been installed and applied in accordance with any relevant recommendations made by the Seller, the Seller will at its option replace the goods or refund the net invoiced price in respect of the goods which have been shown to be defective. If the Seller does so supply substitute goods the Buyer shall be bound to accept such substituted goods in full satisfaction of the obligations of the Seller under the contract.
- (b) The Buyer shall in any event have no claim or off-set in respect of defects unless a written complaint is sent to the Seller as soon as the defect is noticed and no use is made of the goods thereafter or alteration made thereto by the Buyer before the Seller is given an opportunity to inspect the goods.
- (c) The Buyer is responsible for ensuring that the goods are fit for any particular purpose, and no warranty or condition of fitness for any particular purpose is to be implied into the contract.

11 WARRANTY

- (a) The Seller warrants that Products which do not comply with Sections 13 to 15 of the Sale of Goods Act 1979 (as amended) are shown to have been defective at delivery as a result of faulty design workmanship or materials (other than free-issue materials), shall either be repaired or replaced or that, at the Sellers option, a credit or refund for the price thereof shall be given provided always that:
 - (i) the Seller receives written notice of the defect within 12 months of delivery;
 - (ii) no alteration to or interference with the Products takes place before the Seller is given access to the Products to inspect and test the same;
 - (iii) the defect does not consist of a loss shortage or damage to which Clause 9 is expressed to apply;
 - (iv) the defect does not arise by reason of a design specification or instruction given by the Buyer;
 - (v) the Buyer has not defaulted in its obligation to make payment of the contract price for the Products;
 - (vi) the defect shall not be attributable to incorrect storage or use of the Products by the Buyer
 - (vii) the Buyer shall indemnify the Seller in respect of loss or damage arising from any use made of Products after the Buyer became or ought reasonably to have been aware of a defect
- (b) In the event of a valid claim being made in accordance with Clause 11a;
 - (i) the Buyer shall be bound to accept repaired or replacement Products or at the Sellers option credit or repayment and shall not be entitled to terminate the contract.

12 LIABILITY

- (a) The Seller does not exclude liability arising under Section 12 of the Sale of Goods Act 1979 (good title) (as amended) or for death or personal injury caused by its negligence as defined in the Unfair Contract Terms Act 1977, fraudulent misrepresentation or any other type of liability which cannot be excluded or limited.
- (b) Save as provided under Clause 9, 10 and 11. A Seller shall have no liability to the Buyer in connection with or arising from any defect or failure in the Products or otherwise due to the quality, condition, suitability, durability, safety or any aspect or feature of the Products. The Sellers liability, whether in respect of one claim or in the aggregate, shall not exceed the contract price payable under this contract for the supply of Products to be provided under it. The price of the Products is predicted on the basis of the limitation of the Sellers liability is therefore reasonable in all the circumstances. The Buyers agrees that it is its own responsibility to insure adequately to cover any loss or damage in excess of the aforesaid limit of the Sellers liability. Subject to reaching agreement on terms, the Seller and the Buyer may determine an increased level of liability which is to be accepted in Writing by the Seller to cover, in particular specific types of loss or damage which both parties reasonably foresee and anticipate.
- (c) In Clause b the term 'liability' means any form of liability whatsoever including but not limited to liability in misrepresentation and under contract, common law, equality and any statutory provision whether or not based on negligence or breach of any express or implied duty to act with care or skill.
- (d) Notwithstanding any other provisions of these conditions the Buyer shall have no claim against the Seller in respect of any loss.

13 FORCE MAJEURE

Notwithstanding anything herein contained neither the Buyer nor the Seller is to be held liable for any delay or failure to carry out the contract due wholly or in part to an act of God action by any Government British or foreign civil war strikes and/or lockouts where soever occurring fire trade disputes floods or unfavourable weather or any material becoming unavailable or irreplaceable (whether at all or at commercially acceptable prices) or any other circumstances beyond the control of the Seller.

14 INSOLVENCY AND BREACH OF CONTRACT

- In the event that;
 - (a) the Buyer commits any breach of the contract and fails to remedy such breach (if capable of remedy) within a period 30 days from receipt of a notice in writing from the Seller requesting such remedy; or
 - (b) any distress or execution is levied upon any of the goods or property of the Buyer; or
 - (c) the Buyer offers to make any arrangements with or for the benefit of its creditors or (if an individual) becomes subject to a petition for a bankruptcy order or (being a limited company) has a receiver appointed of the whole or any part of its undertaking property or assets; or
- (d) an order is made or a resolution is passed or analogous proceedings are taken for the winding up of the Buyer (save for the purpose of reconstruction or amalgamation with insolvency and previously approved in writing by the Seller) the Seller shall thereupon be entitled without prejudice to its other rights hereunder forthwith to suspend all further deliveries until the default has been made good or to determine the contract and any unfulfilled part thereof or at the Seller's option to make partial deliveries. Notwithstanding any such termination the Buyer shall pay to the Seller at the contract rate for all the goods delivered up to and including the date of termination.

15 INDUSTRIAL PROPERTY RIGHTS

If goods supplied by the Seller to the Buyer's design or specifications infringe or are alleged to infringe any patent or registered design right or copyright the Buyer will indemnify the Seller against all damages, costs and expenses incurred by the Seller as a result of the infringement or allegation. The Buyer will give the Seller all possible help in meeting any infringement claim brought against the Seller.

16 BUYER'S ERROR IN ORDERING

In the event the Buyer orders incorrectly the Seller will be under no obligation to the Buyer to rectify or assist in rectifying the error.

17 COMPANY LITERATURE

The information contained in the advertising, sales, technical and other literature issued by the Seller may be relied upon to be accurate in the exact circumstances in which it is expressed otherwise any illustrations performance details examples of installations and methods of assembly and all other information and data in such literature are based on experience and upon trials under test conditions and are provided for general guidance only. No such information or data shall form part of the contract unless it is specifically referred to in the Quotation.

18 LAW AND JURISDICTION

The contract shall be subject in all respects to English Law and to the jurisdiction of the English Courts.